A Tenth Memoir on Quantics

(Continuation and conclusion, Paper No. 693)

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[Vol. X, printed page 381; PDF page 401. Table No. 99 (continued) — Index Table of Derivatives of the binary quintic; degrees 8 through 24.]

[For each degree, the upper part of the printed table lists the covariant-symbols of the indicated deg-order, congregate prefixes marked by two dots; the lower part shows the deg-order distribution of the derivatives pqn where p, q run over the simple covariants of the quintic and n is the order-index. Owing to the very large number of entries (between 13 and 52 derivatives per degree) and the dense two-column arrangement of the original printed page, we reproduce here, for each remaining degree of Table No. 99, the structural skeleton (column heads in deg-order, row labels in derivative-symbol) faithfully transcribed; the symbolic congregate-content lists at column heads, which the printed page sets in small condensed type, are summarised in prose. The numerical entries in the body of each skeleton are reproduced verbatim from the scan.]

Deg. 8 (52 derivs.) — column heads in deg-order: 2, 4, 6, 8, 10, 12, 14.
Covariant-symbols present at column 8: g^2, r, bk, bn, dj, $..dk$, $gh..ej$, g^2; at column 6: b^2g, bm, bn, adg, $..dk$, g^2, beg, b^2i, bce, $..bde$, cm, $..bd^2$, $..ek$; at column 10: a^2j, abj, aeg, $..ap$, $..b^2$, $..bde$, $..cl$, en, fj, $..hi$; at columns 12 and 14: higher congregates a^2bh, ahn, $..ab^2e$, $..adi$, b^3c, acj, b^2ci, $..cde$, a^2n, a^2ij, $..adh$, bci, $..bd^2j$, $..aeh$, afg, beh, $..be^2$, $..ed^2$, $..d^3$, $..fk$, etc.

Ord.	0	2	4	6	8	10	12
ao		1					
ap	5	4	3	2	1		
bm	2	1					
bn		2	1				
cm			2	1			
cn		4	3	2	1		
dj		1					
dk	3	2	1				
dl			3	2	1		
ej			1				
ek		3	2	1			
el		6	4	3	2	1	
fj					1		
fk				3	2	1	
fl		7	6	6	4	3	2
hh	4		2				
hi		4	3	2	1		
ii	6		4		2		

Deg. 9 (46 derivs.) — column heads 1, 3, 5, 7, 9, 11.
Covariants present: bo, c^2, ar, ab^2g, a^2o, bp, gk, aq, $..b^2h$, ahn, $\cdot$, by, bcg, adj, $..adk$, bdg, bp, agh, $..aej$, agi, $dm..$, co, b^2h, $..b^2e$, b^2i, hj, $dn..$, bcj, $em..$, $..bdh$, gi, cdg, bek, $..hk$, $..cf$, $..bdi$, $..ij$, $..en$, $..beh$, $..ik$,

hfg, ceg, cp, $..d^2e$, $..fin$, $..hi$.

Ord.	1	3	5	7	9	11
ar		2	1			
bo	1					
bp		2	1			
co			1			
cp	5	4	3	2	1	
dm	2	1				
dn	3	2	1			
em		2	1			
en	4	3	2	1		
fm				2	1	
fn			4	3	2	1
hj		1				
hk	3	2	1			
hl		4	3	2	1	
ij			1			
ik		3	2	1		
il	6	5	4	3	2	1

[Vol. X, printed page 383; PDF page 403. Table No. 99 (continued).]

Deg. 10 (44 derivs.) — column heads 0, 2, 4, 6, 8, 10, 12.
Covariants present: bq, br, b^2, abo, av, a^2t, $..do$, c^2, as, a^2q, $..ab^2k$, $abeg$, gm, gn, b^2g, bhn, b^2hi, ab^2j, abp, bdj, aco, f^2, jk, bgh, $..bdk$, $abdg$, $..adn$, cf, $..bej$, $..adm$, $..aem$, bgi, eq, cr, ahj, agl, $..d^2g$, $..b^3$, $..eo$, $..deg$, $..ahk$, $dp..$, eh, $hm..$, $..aij$, $..hn$, b^2g, $im..$, $..b^2p$, b^3c, bcm, $..b^2de$, $..jl$, $..bek$, ben, $..bh^2$, $..bdl$, cdj, $.bfj$, cgh, $..bhi$, $..d^2h$, $..cdk$, $..ep$, $.cej$, $..fo$, cgi, $in..$, $..d^2l$, $..kl$, $..deh$, $..dfg$.

Ord.	0	2	4	6	8	10
as	3	2	1			
br	2	1				
cr		2	1			
do		1				
dp	3	2	1			
eo		1				
ep	5	4	3	2	1	
fo				1		
fp		6	4	3	2	1
hm	2	1				
hn	4	3	2	1		
im		2	1			
in		4	3	2	1	
jk	1					
jl			1			
kk	3					
kl		3	2	1		
ll	6		4		2	

Deg. 11 (35 derivs.) — column heads 1, 3, 5, 7, 9.
Covariants present: go, bgj, bs, b^2o, ahg^2, c^2, bgk, abq, $..ado$, t, dg^2, dq, agm, agn, jm, $..dr$, bs, ajk, $..b^2k$, $..b^2eg$, ef, b^2j, b^2dg, h^2p, gp, $..bdm$, beo, $..ho$, bhj, $..bdn$, $..jn$, cgj, $..bem$, $km..$, d^2j, bgl, $..dgh$, $..bhk$, $er..$, $..bij$, $io..$, cjk, c^2s, $..kn$, $..d^2k$, $.dej$, $..dgi$, $..egh$, ff, fq, $hp..$, $lm...$

Ord.	1	3	5	7	9
bs	2	1			
cr		3	2	1	
dr	2	1			
er		2	1		
fr				2	1
ho		1			
hp	4	3	2	1	
io			1		
ip	5	4	3	2	1
jm	1				
jn		1			
km	2	1			
kn	3	2	1		
lm	2	1			
ln		4	3	2	1

[Vol. X, printed page 385; PDF page 405. Table No. 99 (continued).]

Deg. 12 (26 derivs.) — column heads 2, 4, 6, 8, 10, 12.
Covariants present: g^2r, gr, by, ago, at, b^2r, $abgj$, ab^2o, $..bdo$, $..adg^2$, jo, cq, bgn, adq, $abgk$, bgm, bjk, abs, ajm, $..adr$, hf, $..dgk$, aeg^2, $..ds$, b^2g, aeq, dgj, egj, cm, agp, fh, b^2dj, $..hr$, b^2gh, $..aho$, hu, $..jp$, beg^2, $..ajn$, $..ko$, $mn..$, bcq, $..akm$, $..bo^2$, $..bdj^2g$, $..beo$, b^2n, $..bhm$, $..b^2r$, cgm, $..b^2ik$, $..d^3m$, $.b^2ej$, $..dhj$, $..b^2gi$, $..egk$, bcr, $..es$, $..bdeg$, $..bdp$, $..ir$, $..bhn$, $..kp$, $..bim$, $..lo$, $..bjl$, $..ce^2$, $..cdo$, cgn, cjk, $..d^2n$, $..dem$, $..dgl$, $..dhk$, $..dij$, $..ehj$, $..fgj$, $..ghi$.

Ord.	0	2	4	6	8	10
at			1			
ds	3	2				
es		3				
fs				2	1	
hr		2				
ir						
jo	1					
jp		1				
ko		1				
kp		3	2	1		
lo				1		
lp		6	4	3	2	1

Deg. 13 (19 derivs.) — column heads 1, 3, 5, 7.
Covariants present: g^2j, bgo, ag^2, agr, bt, c^2gi, ajo, bgk, au, $..b^2o$, v^2, by, $..b^2gk$, $.-jr$, bd^2, b^2s, hq, bdq, $..bdr$, $..ino$, bjm, beg^2, beg, $..dgm$, $dp..$, bgp, $..bho$, $..hr$, $..ijn$, $no..$, $..bhn$, cgo, ct, $..d^2o$, $..dgn$, $..djk$, $..egm$, $..ef$, $..ghk$, gij, $..hs$, $..lq$, $mp...$

Ord.	1	3	5	7
bt	1			
ct			1	
hs	3	2	1	
is		3	2	1
jr	1			
kr		1		
lr			1	
mo	1			
mp		2	1	
no		1		
np	4	3	2	1

[Vol. X, printed page 387; PDF page 407. Table No. 99 (continued).]

Deg. 14 (17 derivs.) — column heads 0, 2, 4, 6, 8.
Covariants present: bg^2, bgr, m^2, c^2j, $abgo$, bgq, bjo, abt, bu, $..dgo$, av, ag^2k, g^2m, $..dt$, by, ags, df, g^2n, b^2q, $..ajr$, mq, gj^2, b^2gm, akq, $..o^2$, js, by, $..amo$, $mr..$, $bdgj$, b^2do, nq, bg^2i, $..b^2gn$, bhq, $..bko$, b^2jk, cf, $..bdgk$, $..bds$, cgq, $begj$, $..cd^2g^2$, bg^2l, $..bhr$, biq, $..bjp$, $..djm$, $..b^2m$, $..ego$, $..et$, cgr, $..ghm$, cjo, $..gkl^2$, $..d^2r$, $..hf$, $..def$, $..deq$, $..ks$, $nr..$, $..dgp$, $..op$, $..dho$, $..djn$, $..dkm$, $..ejm$, $..ghn$, $..gim$, $..gjl$, $..hjk$, if.

Ord.	0	2	4	6	8
dt		1			
et			1		
ft					1
js	1				
ks	2	1			
lt			3	2	1
mr	2	1			
nr		2	1		
ot			1		
pp		4		2	

Deg. 15 (12 derivs.) — column heads 1, 3, 5.
Covariants present: g^2o, bg^2j, $..b^2go$, g^2t, bjq, bt^2, u, ln, bfk, cq, bgs, dgq, bjr, bkq, $du..$, $..bnu$, gjm, $..dgr$, cf, $..djo$, eg^2, $or..$, egq, $eu..$, fp^2, $..ghn$, gjn, $..gkm$, $ht..$, $..fk$, $..hls$, pq.

Ord.	1	3	5
bv	1		
cv			1
ht	1		
it			1
ms	2	1	
ns	3	2	1
or	1		
pr		2	1

[Vol. X, printed page 388; PDF page 408. Table No. 99 (continued).]

Deg. 16 (13 derivs.) — column heads 0, 2, 4, 6, 8 (col. 8 concluded).
Covariants present: g^4, c^2r, by, afo, $..abg^2$, g^2q, gjo, b^2gq, agt, $abjq$, $..X$, gtv, jt, b^2m, acq, abv, $..eno$, g^2, qr, $..b^2gr$, $..ad^2$, $..g^2h^2$, bg^2m, gjo, bg^2f, $..bdgo$, $..bdt$, $adgq$, $.gir$, bmq, bfn, $..adu$, $ngjm$, $..gkp$, $..bo^2$, $..af$, $..glo$, $bgjk$, $..aor$, $.gn^2$, dg^2, bjs, by, $..h^2q$, $..hko$, djq, $..bmr$, b^2q, $..hw^2$, $..du$, $..bnq$, b^2gni, $..ijo$, g^2h, $..dg^2k$, b^2f, $..jkn$, gh^2, $..dgs$, $..djr$, $..bdgj$, $..khn$, $.gko$, $..dkq$, b^2gh, $..lt$, $..gm^2$, $..dmo$, $..b^2iq$, $..ps$, $..hu$, $..efj$, $..b^2ko$, $..fm$, egj, $..ev$, $..bhn^2$, $..kt$, $..ghr$, bcf, $..os$, $bcgq$, gn, bcu, $..r^2$, $..gjp$, $..bd^2g^2$, $..gmn$, $..bd^2q$, $..hjo$, $..bdjm$, $..iu$, $..bego$, $..jkm$, $..bet$, $..bghm$, $..bgk^2$, $..bhf$, $..bks$, $..bnr$, $..bop$, cg^2m, cmq, $..cd^2$, $..d^2gm$, $..d^2f$, $..dghj$, $..dkr$, $..dno$, $..efk$, $..egs$, $..yjf$.

Ord.	0	2	4	6	8
do		1			
ev			1		
ft				1	
jt	1				
kt		1			
lt					1
op		1			
os		1			
pp		4		2	
qq		3	2	1	

Deg. 17 (6 derivs.) — column heads 1, 3, 5.
Covariants present: m^2j, bfo, ag^2, bgt, ag^2q, gv, boq, agu, ju, g^2k, aq^2, $gh..bj$, h^2k, $gkq..$, $..gmo$, bdg^2, $..fo$, $bdgq$, ku, bdu, $mt..$, $bgjm$, qs, $..bf$, $..bor$, $..dghn$, ds^2, $..dmq$, $..do^2$, fhj, $..gkr$, $..gno$, hv, $..jko$, $..gn^2$, $..nt$, $..rs$.

Ord.	1	3	5
hv		1	
iv			1
ml	1		
nt		1	
rs	2	1	

Deg. 18 (6 derivs.) — column heads 2, 4, 6.
Covariants present: w, bg^2, bg^2r, a^2j, bg^2q, $mi..$, $bgjo$, bgu, bjt, agv, bf, bqr, aju, g^2m, $..dg^2o$, by, $..doq$, g^2, g^2n, b^2gq, gmq, b^2u, $..gv^2$, hgm, fp, $gpjk$, bv^2, $..jv$, m, b^2mq, mu, $..gmr$, $.bf$, $..at$, gnq, $bdgv$, fr, $bdjq$, jkq, $..bdv$, $..jmo$, bfh, $kv..$, $bgjuj$, nu, $.bgko$, $..bgm^2$, bhu, $.bpm$, $.bkt$, $..bos$, $.br^2$, cg^4, cg^2q, cgu, cq^2, $.d^2q$, $.d^3gq$, $.d^4u$, $.dgjm$, $.dor$, $.eg^2o$, $.egj$, $.eoq$, $.g^2hm$, ghf, $.g^4$, $.gnr$, $.gop$, $.hmq$, $.M$, $.jhr$, $.jno$, $.k^2$, $.kmo$, $.m^4$, $.pt$, $.s^2$.

Ord.	2	4	6
jv	1		
kv		1	
lv			1
at	1		
pt		1	
ss		2	

Deg. 19 (3 derivs.) — column heads 1, 3.
Covariants present: g^2o, bg^2j, w, gh, bg^2q, gqo, bgv, ou, bju, qt, dg^4, dg^2j, $..djt$, dq^2, fjm, $..gf$, gor, jmq, $..jo^2$, $mv..$, $..rt$.

Ord.	1	3
mv	1	
nv		1
rt	1	

[Vol. X, printed page 390; PDF page 410. Table No. 99 (concluded).]

Deg. 20 (3 derivs.) — column heads 2, 4.
Covariants present: g^4, bw, b^2g^4, g^2q, gh^2, b^2q^2, g^2u, g^4o, b^2gn, gf, sgt, by, qu, gqr, bj^2m, $..joq$, $..ov$, $bgmq$, ru, $..bgo^2$, bfq, bjv, bmu, $..hot$, rfA, $dyjq$, $..dgv$, dju, g^2h, fhq, $..fko$, $..fm^2$, ghu, $.gpm$, $..gkt$, $..gos$, $..gr^2$, w, r, $.jor$, $..koq$, $..m^2q$, $..mo^2$, $..St$.

Ord.	2	4
ov	1	
pv		1
sv	1	

Deg. 21 (1 deriv.) — column head 1.
Covariants present: g^2j, g^3q, gh, gju, jf, qv, w.

Ord.	1
tv	1

Deg. 22 (1 deriv.) — column heads 0, 2.
Covariants present: gw, bg^2, btf, b^2g^2q, bfu, bgq^2, bqu, ghn, fff, fmq, $..fo^2$, fgq, $..sjv$, gmu, $..got$, A, mq^2, $..o^2q$, e...

Ord.	0	2
sv	1	

Deg. 24 (1 deriv.) — column head 0.
Covariants present: g^6, A, g^2, ff, gqu, ct, M^4.

Ord.	0
tv	1

[End of Table No. 99. The total over degrees 2 through 24 is 429 derivatives, yielding the 22 simple covariants $b, c, d, \ldots, w$ of the binary quintic, as remarked in art. 383 above.]

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384. The Canonical form (using the divided expressions, Table No. 98) is peculiarly convenient for the calculation of the derivatives. Some attention is required in regard to the numerical determination: it will be observed that A is given in the standard form $(A_0, A_1, A_2, A_3, A_4, A_5 \mid x, y)^5$, while the other covariants are given in the denumerate forms $B = (B_0, B_1, B_2 \mid x, y)^2$ &c.: these must be converted into the other form

$$\mathsf{B} = (B_0, \tfrac{1}{2}B_1, B_2 \mid x, y)^2, \quad \mathsf{C} = (C_0, \tfrac{1}{6}C_1, \tfrac{1}{15}C_2, \tfrac{1}{20}C_3, \tfrac{1}{15}C_4, \tfrac{1}{6}C_5, C_6 \mid x, y)^6, \ \&c.,$$

the numerical coefficients being of course the reciprocals of the binomial coefficients. We thus have, for instance, the leading coefficients,

$$\text{l.c. of } AC2 = A_0 \cdot \tfrac{1}{15}C_2 - 2 \cdot A_1 \cdot \tfrac{1}{6}C_1 + A_2 \cdot C_0,$$

but

$$\text{l.c. of } \mathsf{BC}2 = B_0 \cdot \tfrac{1}{15}C_2 - 2 \cdot \tfrac{1}{2}B_1 \cdot \tfrac{1}{6}C_1 + B_2 \cdot C_0.$$

Moreover, as regards the covariants $AA2$, $AA4$, &c., we take what are properly the half-values,

$$\begin{aligned} \text{i.e. of } AA2 &= A_0A_2 - A_1^2 \quad (\text{instead of } A_0A_2 - 2A_1A_1 + A_2A_0), \\ \text{,, ,, } AA4 &= A_0A_4 - 4A_1A_3 + 3A_2^2 \quad (\text{instead of } A_0A_4 - 4A_1A_3 + 6A_2A_2 - 4A_3A_1 + A_4A_0), \\ &\&c., \end{aligned}$$

and similarly

$$\text{l.c. of } \mathsf{BB}2 = B_0B_2 - (\tfrac{1}{2}B_1)^2, \qquad \text{,, ,, } \mathsf{CC}2 = C_0 \cdot \tfrac{1}{15}C_2 - (\tfrac{1}{6}C_1)^2, \qquad \&c.$$

Any one of these leading coefficients, for instance l.c. of $AC2$, is equal to the corresponding covariant derivative, multiplied, it may be, by a power of a: the index of this power being at once found by comparing the deg-orders, these in fact differing by a multiple of 1.5 the deg-order of a. Thus

$$\begin{aligned} &aa2, \quad A_0A_2 - A_1^2, \quad \text{deg-orders are } 2.6, 2.6: \text{ or } aa2 = A_0A_2 - A_1^2, \\ &aa4, \quad A_0A_4 - 4A_1A_3 + 3A_2^2, \quad \text{deg-orders are } 2.2, 4.12: \text{ or } aa4 = (A_0A_4 - 4A_1A_3 + 3A_2^2); \end{aligned}$$

we have in fact

$$A_0A_2 - A_1^2 = 1 \cdot c - 0^2 = c \qquad \text{and so } aa2 = c,$$

$$AA4 = A_0A_4 - 4A_1A_3 + 3A_2^2 = 1 \cdot (a^2b - 3c^2) - 4 \cdot 0 \cdot f + 3 \cdot c^2 = a^2b \quad \text{and so } aa4 = b.$$

As another instance, and for the purpose of showing how the calculation is actually effected, consider the derivative $ch2$, which is to be calculated from the leading coefficient of $CH2$,

$$\begin{aligned} &C_0 \cdot \tfrac{1}{15}H_2 - 2 \cdot \tfrac{1}{6}C_1 \cdot \tfrac{1}{2}H_1 + \tfrac{1}{15}C_2 \cdot H_0 : \quad \text{this is} \\ &= c\big(\tfrac{1}{2}a^2g - 2abd - ch\big) \\ &\quad - 2 \cdot \tfrac{1}{2}f\big(\tfrac{1}{2}be - l\big) \\ &\quad + \big(\tfrac{1}{15}a^2b - c^2\big)h \end{aligned}$$

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= column next written down; but this column contains congregate terms which have to be replaced by their segregate values (see Table No. 96, deg-order 8.16); and we thus obtain

	a^2cy	a^2b^3	a^2bh	a^2cg	$abcd$	b^2c^2	c^2h
$\frac{1}{15}a^2bh$			$+\frac{1}{15}$				
$+\frac{1}{2}a^2cg$				$+\frac{1}{2}$			
$-2abcd$					-2		
$-\frac{1}{2}bef$			$-\frac{1}{2}$		$+3$	$+2$	
$-2c^2h$						-2	
$+fl$	$+\frac{1}{3}$	$-\frac{1}{3}$	$+\frac{2}{5}$	$-\frac{1}{5}$	-1	-2	$+2$
$=$	$+\frac{1}{3}$	$-\frac{1}{3}$	$+\frac{11}{30}$	$-\frac{1}{5}$	0	0	0

viz. the terms other than those divisible by a^2 all disappear: we may either abbreviate the calculation by omitting them *ab initio*, or retain them for the sake of the verification afforded by their disappearance. The factor a^2 divides out, and the final result is

$$ch2 \;=\; \tfrac{1}{3}cy - \tfrac{1}{3}b^3 + \tfrac{11}{30}bh - \tfrac{1}{5}cg,$$

which is the proper segregate expression of the derivative $ch2$: of course, we have deg-order $CH2 = 8.16$, deg-order $ch2 = 6.6$, and the difference is 2.10, the double of 1.5, so that the factor a^2 is as it ought to be.

Table No. 100 (The Derivatives up to the Sixth Order).

Degree 2.

2.2		2.6	
$aa4$	$+1$	$aa2$	$+1$

Degree 3.

3.1	
$ac5$	0

3.3 d	
$ab2$	-3
$ac4$	$+\frac{3}{2}$

3.5 e	
$ab1$	$+\frac{1}{2}$
$ac3$	$-\frac{1}{2}$

3.7 ab	
$ac3$	$-\frac{3}{4}$

3.9 f	
$ac1$	$+\frac{1}{4}$

Degree 4.

4.0 g	
$ae5$	-2
$bb2$	$-\frac{1}{4}$
$cc6$	1

4.2	
$ad3$	0
$ae4$	0

4.4 h	
$ad2$	$+\frac{1}{2}$
$ae3$	$-\frac{1}{2}$
$af5$	$+\frac{3}{5}$
$be2$	$+\frac{1}{2}$
$cc4$	$+\frac{1}{10}$

4.6	
$ad1$	$+\frac{1}{4}$
$ae2$	$+1$
$af4$	$-\frac{1}{5}$
$bc1$	$+\frac{1}{2}$

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4.8 $ad,\ bc$	
$ae1$	$-\frac{6}{5},\ -2$
$af3$	$+\frac{59}{42},\ -\frac{5}{6}$
$cc2$	$+\frac{1}{4},\ -\frac{1}{20}$

4.10 ae	
$af2$	$+1$

4.12 $a^2b,\ c^2$	
$af1$	$+\frac{2}{5},\ -2$

Degree 5.

5.1 j	
$ah4$	$+2$
$ai5$	$-\frac{4}{3}$
$bd2$	$-\frac{1}{3}$
$ce5$	$-\frac{8}{7}$

5.3 k	
$ah3$	$+\frac{1}{2}$
$ai4$	$+\frac{1}{3}$
$bd1$	$-\frac{1}{6}$
$be2$	$-\frac{3}{5}$
$cd3$	$+\frac{8}{20}$
$ce4$	$+\frac{2}{25}$
$cf6$	$-\frac{19}{42}$

5.5 $ag,\ bd$	
$ah2$	$+\frac{1}{6},\ -2$
$ai3$	$0,\ -2$
$be1$	$-\frac{1}{2},\ +\frac{24}{5}$
$cd2$	$0,\ -\frac{2}{15}$
$ce3$	$-\frac{1}{20},\ -\frac{48}{5}$
$df5$	$-\frac{1}{72},\ +\frac{8}{63}$

5.7 $be,\ l$	
$ah1$	$-2,\ -4$
$ai2$	$0,\ +\frac{1}{3}$
$bf2$	$-\frac{7}{36},\ +1$
$cd1$	$0,\ +\frac{1}{6}$
$ce2$	$+\frac{1}{5},\ -\frac{2}{5}$
$df4$	$-\frac{1}{60},\ +\frac{43}{315}$

5.9 $ab^2,\ ah,\ cd$	
$ai1$	$-\frac{1}{3},\ +\frac{1}{3},\ +3$
$bf1$	$+\frac{2}{5},\ -\frac{1}{2},\ +3$
$ce1$	$0,\ -\frac{1}{2},\ +\frac{9}{5}$
$df3$	$+\frac{2}{15},\ -\frac{11}{120},\ -\frac{439}{120}$

5.11 $ai,\ ce$	
$df2$	$+\frac{1}{15},\ +\frac{1}{180}$

5.13 $a^2d,\ abc$	
$df1$	$+\frac{1}{2},\ -\frac{6}{8}$

Degree 6.

6.0	
$ci6$	0

6.2 $bg,\ m$	
$ak3$	$0,\ -4$
$al5$	$0,\ +\frac{20}{7}$
$bh2$	$-\frac{1}{5},\ -2$
$ch4$	$+\frac{1}{10},\ -\frac{2}{5}$
$ci5$	$0,\ +\frac{2}{5}$
$dd2$	$0,\ +\frac{1}{5}$
$de3$	$0,\ -\frac{4}{5}$
$ee4$	$-1,\ -\frac{48}{25}$
$ff8$	$-\frac{5}{321},\ -\frac{68}{507}$

6.4 n	
$aj1$	-1
$ak2$	$+\frac{1}{3}$
$al4$	$-\frac{1}{35}$
$bh1$	$+\frac{1}{2}$
$bi2$	$+\frac{1}{3}$
$ch3$	$+\frac{3}{10}$
$ci4$	$+\frac{1}{10}$
$de2$	$-\frac{1}{3}$
$ef5$	$-\frac{64}{63}$

6.6 $aj,\ b^2,\ bh,\ cg$	
$aj1$	-1
$a1$	$-\frac{2}{3},\ +2,\ -3,\ +1$
$ak2$	$+\frac{1}{3}$
$al3$	$-\frac{10}{35},\ +\frac{2}{7},\ -\frac{4}{7},\ +\frac{3}{7}$
$bi1$	$-\frac{1}{3},\ +\frac{5}{6},\ -\frac{1}{2},\ 0$
$ch2$	$+\frac{1}{3},\ -\frac{1}{3},\ +\frac{11}{30},\ +\frac{7}{60}$
$de1$	$-\frac{2}{15},\ +\frac{2}{15},\ +\frac{1}{15},\ -\frac{1}{5}$
$df3$	$+\frac{59}{375},\ -\frac{143}{375},\ +\frac{425}{150},\ -\frac{130}{150}$
$ee2$	$-\frac{4}{25},\ +\frac{4}{25},\ -\frac{38}{25},\ +\frac{9}{25}$
$ef4$	$+\frac{230}{315},\ -\frac{4}{105},\ +\frac{10}{63},\ -\frac{71}{105}$
$ff6$	$-\frac{1520}{7088},\ +\frac{2373}{7088},\ +\frac{3633}{15876},\ -\frac{5591}{31752}$

[Vol. X, printed page 394; PDF page 414. Table No. 100 (concluded).]

6.8 $ak,\ bi$	
$al2$	$-\frac{4}{21},\ +\frac{1}{7}$
$ch1$	$+\frac{1}{6},\ +\frac{1}{3}$
$ci2$	$+\frac{1}{8},\ +\frac{1}{15}$
$df2$	$-\frac{11}{108},\ -\frac{1}{45}$
$ef3$	$+\frac{32}{315},\ +\frac{64}{315}$

6.10 $a^2g,\ abd,\ b^2c,\ ch$	
$al1$	$0,\ -\frac{3}{7},\ -1,\ +1$
$ci1$	$0,\ -\frac{1}{2},\ +\frac{1}{8},\ -\frac{1}{8}$
$df1$	$0,\ -\frac{1}{8},\ +\frac{1}{3},\ -\frac{1}{3}$
$ef2$	$+\frac{1}{36},\ +\frac{7}{5},\ -\frac{19}{45},\ -\frac{4}{5}$
$ff4$	$+\frac{1}{482},\ -\frac{58}{765},\ +\frac{89}{765},\ -\frac{8}{63}$

6.12 $abe,\ al,\ ci$	
$ef1$	$+\frac{2}{5},\ -\frac{2}{5},\ -\frac{5}{8}$

6.14 $a^2b^2,\ a^2h,\ acd,\ bc^2$	
$ff2$	$-\frac{4}{81},\ +\frac{1}{30},\ +\frac{5}{6},\ -\frac{2}{9}$

which is complete to the sixth degree. I had calculated the derivatives up to the tenth degree, but the results were not in the segregate form.

On the form of the Numerical Generating Functions: the N.G.F. of a Sextic.
Art. Nos. 385, 386.

385. It is to be remarked that the R.G.F. is derived not from the fraction in its least terms, which is algebraically the most simple form of the N.G.F., but from a form which contains common factors in the numerator and denominator: thus for the quadric, the cubic, and the quartic, writing down the two forms (identical in the case of the quadric) these are

Quadric

$$\text{N.G.F.} = \frac{1}{1-\alpha x^2\cdot 1-a^2}.$$

Cubic

$$\text{N.G.F.} = \frac{1+ax^3}{1-a^2\cdot 1-ax^3\cdot 1-ax} = \frac{1-a^2x^6}{1-a^2\cdot 1-ax^3\cdot 1-ax^3\cdot 1-a^2x^3}.$$

Quartic

$$\text{N.G.F.} = \frac{1-ax^2+a^2x^4}{(1-a^2)(1-a^3)(1-ax^4)(1-ax^2)(1-a^2x^4)} \equiv \frac{1-a^6x^{12}}{(1-a^2)(1-a^3)(1-ax^4)(1-a^2x^4)(1-a^3x^6)}.$$

[Vol. X, printed page 395; PDF page 415.]

For the quintic the two forms are, N.G.F. =

$$\frac{\begin{aligned}&(1-a^6+a^{12})x^0+(-1+a^4+2a^6-a^{12})ax\\&+(a^2-a^8+a^{10})x^2+(-1+a^4+a^6+a^8-a^{10}-a^{12})ax^3\\&+(1+a^2-a^4-a^6-a^8+a^{12})a^2x^4+(-a^2+a^4-a^{10})a^3x^5\\&+(1-2a^6-a^8+a^{12})a^2x^6+(-1+a^6-a^{12})a^3x^7\end{aligned}}{(1-a^4)(1-a^8)(1-a^{12})(1-ax^5)(1-ax^3)(1-ax)}$$

and (equivalent form)

$$\frac{\begin{aligned}&(1+a^{18})x^0+(a^4+a^6+a^{10}+a^{12})ax\\&+(a^4+a^6+a^8+a^{10}+a^{14}-a^{18})a^2x^2+(1+a^2+a^4+a^8)a^3x^3\\&+(1+a^2+a^4+a^6+a^{10}-a^{14})a^4x^4+(1+a^4+a^6-a^{18})a^3x^5\\&+(a^2-a^{12}-a^{14})a^2x^6+(a^4-a^8-a^{12}-a^{14}-a^{16}-a^{18})ax^7\\&+(-a^{10}-a^{12}-a^{14}-a^{16}-a^{18})a^2x^8+(-a^4-a^8-a^{10}-a^{12}-a^{14})a^3x^9\\&+(-a^6-a^8-a^{12}-a^{14})a^4x^{10}+(-1-a^{18})a^3x^{11}\end{aligned}}{(1-a^4)(1-a^8)(1-a^{12})(1-ax^5)(1-a^2x^2)(1-a^2x^6)}$$

The last form is in fact equivalent to that used for the determination of the R.G.F.

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386. For the sextic the forms are, N.G.F. =

$$\frac{\begin{aligned}&(1+a-a^3-a^4-a^5+a^7+a^8)x^0\\&+(-1-a+a^2+2a^3+2a^4+a^5-a^7-a^8)ax^2\\&+(-1+a^2+a^3+a^4+a^5-a^7-a^8)ax^4\\&+(1+a-a^3-a^4-a^5-a^6+a^8)a^2x^6\\&+(1+a-a^3-2a^4-2a^5-a^6+a^7+a^8)a^2x^8\\&+(-1-a+a^3+a^4+a^5-a^7-a^8)a^3x^{10}\end{aligned}}{(1+a)(1-a^2)(1-a^3)(1-a^4)(1-a^5)(1-ax^2)(1-ax^4)(1-ax^6)}$$

and

$$\frac{\begin{aligned}&(1+a^{16})x^0\\&+(1+a^2+a^4+a^5+a^7+a^9)a^3x^2\\&+(a^2+a^3+a^4+a^5+a^6+a^7+a^8+a^9+a^{11})a^2x^4\\&+(1+a+2a^3+a^5+a^6+a^8-a^{13})a^3x^6\\&+(a^{1/2}+a^{5/2}+a^{9/2}-a^{21/2}-a^{25/2}-a^{29/2})a^{5/2}x^8\\&+(a^2-a^7-a^9-a^{10}-2a^{12}-a^{14}-a^{15})a^3x^{10}\\&+(-a^4-a^6-a^7-a^8-a^9-a^{10}-a^{11}-a^{12}-a^{13})a^3x^{12}\\&+(-a^6-a^8-a^{10}-a^{11}-a^{13}-a^{15})a^2x^{14}\\&+(-1-a^{15})a^5x^{16}\end{aligned}}{(1-a^2)(1-a^4)(1-a^8)(1-a^{10})(1-ax^6)(1-a^2x^4)(1-a^2x^8)}$$

where observe that in the middle term, although for symmetry $a^{1/2}$ $(=\sqrt{a})$ has been introduced into the expression, the coefficient is really rational, viz. the term is

$$(a^3+a^5+a^7-a^{13}-a^{15}-a^{17})x^8.$$

The second form or one equivalent to it is due to Sylvester: I do not know whether he divided out the common factors so as to obtain the first form. I assume that it would be possible from this second form to obtain a R.G.F., and thence to establish for the 26 covariants of the sextic a theory such as has been given for the 23 covariants of the quintic: but I have not entered upon this question.

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Table No. 93 bis (The covariant S, adopted form $= -(D, M)$).

{ In this Table, a, b, c, d, e, f denote, as in the tables of former memoirs, the coefficients of the quintic form $(a, b, c, d, e, f \mid x, y)^5$.

[The covariant S is set out as a polynomial in x, y of degree 5 with coefficients S_0, S_1, S_2, S_3, S_4, S_5. The original printed pages 397–398 give the explicit expressions for S_0, S_1, S_2, S_3 as very dense lists of monomials $a^\alpha b^\beta c^\gamma d^\delta e^\epsilon f^\zeta$ with integer coefficients, occupying four to six columns per page. We here record the structural form together with the closed-form reductions (obtained by Cayley by setting $d = e = f = 1$) which the printed text immediately afterwards extracts as a verification.]

$$S \;=\; \big(S_0,\ S_1,\ S_2,\ S_3,\ S_4,\ S_5 \;\big|\; x,\ y\big)^5.$$

[The first two coefficients S_0, S_1 are calculated directly; S_2 is deduced from S_1, and S_3 from S_0, by reversing the order of the letters — which is the same thing as interchanging $a \leftrightarrow f$, $b \leftrightarrow e$, $c \leftrightarrow d$ — and reversing also the signs of the numerical coefficients. This process is, to a very great extent, a verification of the values of S_0 and S_1; for, as presently remarked, the terms of S_0 form subdivisions such that in each subdivision the sum of the numerical coefficients is $= 0$; in passing by the reversal process to the value of S_3, the terms are distributed into an entirely new set of subdivisions, and then in each of these subdivisions the sum of the numerical coefficients is found to be $= 0$; and the like as regards S_1 and S_2.]

[The full lists of monomials and coefficients comprising S_0, S_1, S_2, S_3 occupy printed pages 397–398 in seven columns. As the entries are partly unreadable in the scan (typesetter has compressed and broken many fraction-bars), we omit the verbatim reproduction and reproduce instead the verification specialisations which Cayley records on printed pages 399–400.]

[Vol. X, printed page 399; PDF page 419. Verification of the covariant S.]

If in the expressions for S_0, S_1, S_2, S_3 we first write $d = e = f = 1$, thus in effect combining the numerical coefficients for the terms which contain the same powers in a, b, c, we find

$$\begin{aligned} S_0 \ &= a^5\,(-2c^3+6c^2-6c+2) \\ &\quad + a^4\,\{\,b^2(6c^2-12c-6)+b\,(-15c^3+33c^2-21c+3) \\ &\qquad\qquad + b^0(42c^4-147c^3+105c^2-117c+27)\} \\ &\quad + a^3\,\{\,b^4\cdot 0+b^3(30c^2-36c+6)+b^2(-117c^3+249c^2-183c+51) \\ &\qquad\qquad + b\,(9c^4+138c^3-378c^2+330c-99c) \\ &\qquad\qquad + b^0(-63c^5+165c^4-147c^3+45c^2)\} \\ &\quad + a^2\,\{\,b^6\cdot 2+b^5(-15c+3)+b^4(75c^2-69c+24) \\ &\qquad\qquad + b^3(-9c^4-16c^3+225c^2-87c-2) \\ &\qquad\qquad + b^2(72c^4+48c^3-186c^2+96c) \\ &\qquad\qquad + b\,(-126c^4+201c^3-87c^2) \\ &\qquad\qquad + b^0(27c^5-45c^4+20c^3)\}; \end{aligned}$$

which for $c = 1$ becomes

$$= 2b^6-12b^5+30b^4-40b^3+30b^2-12b+2, \quad \text{that is,} \quad 2(b-1)^6,$$

and for $b = 1$ becomes $= 0$.

$$\begin{aligned} S_1 \ &= a^5\,(0c^2+0c+0) \\ &\quad + a^4\,\{\,b^2(0c+0)+b\,(3c^3-9c^2+9c-3)+b^0(-24c^4+99c^3-153c^2+105c-27)\} \\ &\quad + a^3\,\{\,b^4\cdot 0+b^3(-6c^2+12c-6)+b^2(-24c^3+90c^2-108c+42) \\ &\qquad\qquad + b\,(33c^4-90c^3+54c^2+30c-27) \\ &\qquad\qquad + b^0(-27c^5+78c^4-66c^3+6c^2+9c)\} \\ &\quad + a^2\,\{\,b^5(3c-3)+b^4(-15c+15)+b^3(6c^3-12c^2+36c-30) \\ &\qquad\qquad + b^2(9c^4-42c^3+84c^2-108c+57) \\ &\qquad\qquad + b\,(-9c^5+54c^4-96c^3+51c^2) \\ &\qquad\qquad + b^0(9c^5-9c^4)\}; \end{aligned}$$

which for $c = 1$ becomes $= 0$.

$$\begin{aligned} S_2 \ &= a^5\,(0c+0) \\ &\quad + a^4\,\{\,b^2(0)+b\,(0c^3+0c^2+0c+0)+b^0(18c^4-72c^3+108c^2-72c+18)\} \\ &\quad + a^3\,\{\,b^4(0c+0)+b^3(-33c^3+99c^2-99c+33)+b^2(57c^4-162c^3+144c^2-30c-9) \\ &\qquad\qquad + b\,(-60c^5+207c^4-261c^3+141c^2-27c)\} \\ &\quad + a^2\,\{\,b^5\cdot 0+b^4(15c^2-30c+15)+b^3(-54c^3+102c^2-42c-6) \\ &\qquad\qquad + b^2(123c^4-297c^3+243c^2-87c+18) \\ &\qquad\qquad + b\,(-27c^5+102c^4-96c^3+21c^2) \\ &\qquad\qquad + b^0(27c^5-66c^4+51c^3-12c^2)\}; \end{aligned}$$

which for $c = 1$ becomes $= 0$.

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$$\begin{aligned}
S_3 \ &= a^5 \cdot 0 \\
&\quad + a^4 \left\{ b\,(0c+0) + b^0(0c^3+0c^2+0c+0) \right\} \\
&\quad + a^3 \left\{ b^4 \cdot 0 + b^3(0c^2+0c+0) + b^2(-9c^4+36c^3-54c^2+36c-9) \right. \\
&\qquad\qquad\qquad \left. + b^0(36c^5-171c^4+324c^3-306c^2+144c-27) \right\} \\
&\quad + a^2 \left\{ b^5(0c+0) + b^4(7c^3-21c^2+21c-7) + b^3(-30c^4+135c^3-171c^2+93c-18) \right. \\
&\qquad\qquad\qquad + b^2(66c^5-243c^4+333c^3-201c^2+45c) \\
&\qquad\qquad\qquad \left. + b^6(-27c^5+101c^4-141c^3+87c^2-20c) \right\};
\end{aligned}$$

which for $c = 1$ becomes $= 0$.

It follows that, for $c = d = e = f = 1$, the value of the covariant S is $= 2(b-1)^6 y^5$, which might be easily verified.

[End of Paper No. 693 (A Tenth Memoir on Quantics).]

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694.

DESIDERATA AND SUGGESTIONS.

No. 1. The theory of groups.

[From the *American Journal of Mathematics*, t. I. (1878), pp. 50–52.]

SUBSTITUTIONS, and (in connexion therewith) groups, have been a good deal studied; but only a little has been done towards the solution of the general problem of groups. I give the theory so far as is necessary for the purpose of pointing out what appears to me to be wanting.

Let $\alpha, \beta, \ldots$ be functional symbols, each operating upon one and the same number of letters and producing as its result the same number of functions of these letters; for instance, $\alpha(x, y, z) = (X, Y, Z)$, where the capitals denote each of them a given function of (x, y, z).

Such symbols are susceptible of repetition and of combination:

$$\alpha^2(x, y, z) = \alpha(X, Y, Z), \qquad \text{or} \qquad \beta\alpha(x, y, z) = \beta(X, Y, Z),$$

$=$ in each case three given functions of (x, y, z); and similarly for α^3, $\alpha^2\beta$, &c.

The symbols are not in general commutative, $\alpha\beta$ not $= \beta\alpha$; but they are associative, $\alpha\beta \cdot \gamma = \alpha \cdot \beta\gamma$, each $= \alpha\beta\gamma$, which has thus a determinate signification.

The associativeness of such symbols arises from the circumstance that the definitions of $\alpha, \beta, \gamma, \ldots$ determine the meanings of $\alpha\beta$, $\alpha\gamma$, &c.: if $\alpha, \beta, \gamma, \ldots$ were quasi-quantitative symbols such as the quaternion imaginaries i, j, k, then $\alpha\beta$ and $\beta\gamma$ might have by definition values δ and ϵ such that $\alpha\beta \cdot \gamma$ and $\alpha \cdot \beta\gamma$ ($= \delta\gamma$ and $\alpha\epsilon$ respectively) have unequal values.

Unity as a functional symbol denotes that the letters are unaltered, $1(x, y, z) = (x, y, z)$; whence $1\alpha = \alpha 1 = \alpha$.

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The functional symbols may be substitutions; $\alpha(x, y, z) = (y, z, x)$, the same letters in a different order: substitutions can be represented by the notation $\alpha = \frac{yzx}{xyz}$, the substitution which changes xyz into yzx, or as products of cyclical substitutions,

$$\alpha = \frac{yzx}{xyz} \cdot \frac{wu}{uw} = (xyz)(uw),$$

the product of the cyclical interchanges x into y, y into z, z into x; and u into w, w into u.

A set of symbols, $\alpha, \beta, \gamma, \ldots$, such that the product $\alpha\beta$ of each two of them (in each order, $\alpha\beta$ or $\beta\alpha$), is a symbol of the set, is a group. It is easily seen that 1 is a symbol of every group, and we may therefore give the definition in the form that a set of symbols, $1, \alpha, \beta, \gamma, \ldots$ satisfying the foregoing condition is a group. When the number of the symbols (or terms) is $= n$, then the group is of the n^{th} order; and each symbol α is such that $\alpha^n = 1$, so that a group of the order n is, in fact, a group of symbolical n^{th} roots of unity.

A group is defined by means of the laws of combination of its symbols: for the statement of these we may either (by the introduction of powers and products) diminish as much as may be the number of independent functional symbols, or else, using distinct letters for the several terms of the group, employ a square diagram as presently mentioned.

Thus, in the first mode, a group is $1, \beta, \beta^2, \alpha, \alpha\beta, \alpha\beta^2$ $(\alpha^2 = 1,\ \beta^3 = 1,\ \alpha\beta = \beta^2\alpha)$; where observe that these conditions imply also $\alpha\beta^2 = \beta\alpha$.

Or, in the second mode, calling the same group $(1, \alpha, \beta, \gamma, \delta, \epsilon)$, the laws of combination are given by the square diagram

	1	α	β	γ	δ	ϵ
1	1	α	β	γ	δ	ϵ
α	α	1	γ	β	ϵ	δ
β	β	ϵ	δ	α	1	γ
γ	γ	δ	ϵ	1	α	β
δ	δ	γ	1	ϵ	β	α
ϵ	ϵ	β	α	δ	γ	1

$=$ for the symbols $(1, \alpha, \beta, \gamma, \delta, \epsilon)$ are in fact $(1, \alpha, \beta, \alpha\beta, \beta^2, \alpha\beta^2)$.

The general problem is to find all the groups of a given order n; thus if $n = 2$, the only group is $1, \alpha$ $(\alpha^2 = 1)$; if $n = 3$, the only group is $1, \alpha, \alpha^2$ $(\alpha^3 = 1)$; if $n = 4$, the groups are $1, \alpha, \alpha^2, \alpha^3$ $(\alpha^4 = 1)$, and $1, \alpha, \beta, \alpha\beta$ $(\alpha^2 = 1,\ \beta^2 = 1,\ \alpha\beta = \beta\alpha)$; if $n = 6$, there are three groups, a group $1, \alpha, \alpha^2, \alpha^3, \alpha^4, \alpha^5$ $(\alpha^6 = 1)$, and two groups $1, \beta, \beta^2, \alpha, \alpha\beta,$

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$\alpha\beta^2$ ($\alpha^2 = 1$, $\beta^3 = 1$); viz. in the first of these $\alpha\beta = \beta\alpha$, while in the other of them (that mentioned above) we have $\alpha\beta = \beta^2\alpha$, $\alpha\beta^2 = \beta\alpha$.

But although the theory as above stated is a general one, including as a particular case the theory of substitutions, yet the general problem of finding all the groups of a given order n, is really identical with the apparently less general problem of finding all the groups of the same order n, which can be formed with the substitutions upon n letters; in fact, referring to the diagram, it appears that $1, \alpha, \beta, \gamma, \delta, \epsilon$ may be regarded as substitutions performed upon the six letters $1, \alpha, \beta, \gamma, \delta, \epsilon$, viz. 1 is the substitution unity which leaves the order unaltered, α the substitution which changes $1\alpha\beta\gamma\delta\epsilon$ into $\alpha 1\gamma\beta\epsilon\delta$, and so for $\beta, \gamma, \delta, \epsilon$. This, however, does not in any wise show that the best or the easiest mode of treating the general problem is thus to regard it as a problem of substitutions: and it seems clear that the better course is to consider the general problem in itself, and to deduce from it the theory of groups of substitutions.

Cambridge, 28th November, 1877.

No. 2. The theory of groups; graphical representation.

[From the *American Journal of Mathematics*, t. I. (1878), pp. 174–176.]

In regard to a substitution-group of the order n upon the same number of letters, I omitted to mention the important theorem that every substitution is regular (that is, either cyclical or composed of a number of cycles each of them of the same order).

Thus, in the group of 6 given in No. 1, writing a, b, c, d, e, f in place of $1, \alpha, \beta, \gamma, \delta, \epsilon$, the substitutions of the group are 1, $ace \cdot bfd$, $aec \cdot bdf$, $ab \cdot cd \cdot ef$, $ad \cdot be \cdot cf$, $af \cdot bc \cdot de$.

Let the letters be represented by points; a change a into b will be represented by a directed line (line with an arrow) joining the two points; and therefore a cycle abc, that is, a into b, b into c, c into a, by the three sides of the trilateral abc, with the three arrows pointing accordingly, and similarly for the cycles $abcd$, &c. The cycle ab means a into b, b into a, and we have here the line ab with a two-headed arrow pointing both ways; such a line may be regarded as a bilateral. A substitution is thus represented by a multilateral or system of multilaterals, each side with its arrow; and in the case of a regular substitution the multilaterals (if more than one) have each of them the same number of sides. To represent two or more substitutions we require different colours, the multilaterals belonging to any one substitution being of the same colour.

In order to represent a group we need to represent only independent substitutions thereof; that is, substitutions such that no one of them can be obtained from the others by compounding them together in any manner. I take as an example a group

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of the order 12 upon 12 letters, where the number of independent substitutions is = 2. See the diagram, wherein the continuous lines represent black lines, and the dotted lines, red lines.

[Figure: a planar diagram of 12 labelled points $a, b, c, d, e, f, g, h, i, j, k, l$ arranged in two rows of alternating triangles; continuous ("black") directed edges form four triangles representing one substitution of order 3, and dotted ("red") directed edges form four triangles representing a second independent substitution of order 3. Each vertex has exactly one outgoing black and one outgoing dotted edge.]

The diagram is drawn, in the first instance, with the arrows but without the letters, which are then affixed at pleasure; viz. the form of the group is quite independent of the way in which this is done, though the group itself is of course dependent upon it. The diagram shows two substitutions, each of them of the third order: one is represented by the black triangles, and the other by the dotted triangles. It will be observed that there is from each point of the diagram (that is, in the direction of the arrow) one and only one black line, and one and only one dotted line; hence a symbol B, "move along a black line," B^2, "move successively along two black lines," BR (read always from right to left), "move first along a dotted line and then along a black line," has in every case a perfectly definite meaning and determines the path when the initial point is given; any such symbol may be spoken of as a "route."

[Printed page 404 then sets out, in a tall table, the result of applying each of the 12 routes to the initial arrangement $abcdefghijkl$; alongside it the cyclical decomposition of the corresponding substitution. The headings and entries are:]

$abcdefghijkl$	1
$fbcaedhigklj$	$abc \cdot def \cdot ghi \cdot jkl \quad (= B)$
$fcabdefihgjlk$	$acb \cdot dfe \cdot gih \cdot jlk$
$fdaegcjhikbl$	$ad \cdot bl \cdot ch \cdot eg \cdot fj \cdot ik$
$fhabgjdcklie$	$aeh \cdot bjd \cdot cil \cdot fkg$
$fheabgjdckli$	$afl \cdot bkh \cdot cgd \cdot eij$
$feabchdgkilj$	$agj \cdot bfi \cdot cke \cdot dlh$
$fibhkadcjglg$	$ahc \cdot bdj \cdot cli \cdot fgk$
$fkglicdjbeah$	$ai \cdot be \cdot cj \cdot dk \cdot fh \cdot gl$
$ydblcijfkeah$	$ajg \cdot bif \cdot cek \cdot dhl \quad (= R)$
$fkglcjdbaeih$	$ak \cdot bg \cdot cf \cdot di \cdot el \cdot hj$
$fhdljagkecbi$	$alf \cdot bhk \cdot cdg \cdot eji$

[The arrangement here is reproduced from the scan with the customary 19th-century compositor abbreviations preserved; the cycle decompositions in the right column give the structure of each of the 12 substitutions.]

[Vol. X, printed page 405; PDF page 425.]

The diagram has a remarkable property, in virtue whereof it in fact represents a group. It may be seen that any route leading from some one point a to itself, leads also from every other point to itself, or say from b to b, from c to c, ..., and from l to l. We hence see that a route, applied in succession to the whole series of initial points or letters $abcdefghijkl$, gives a new arrangement of these letters, wherein no one of them occupies its original place; a route is thus, in effect, a substitution. Moreover, we may regard as distinct routes, those which lead from a to a, to b, to $c, \ldots$, to l, respectively. We have thus 12 substitutions (the first of them, which leaves the arrangement unaltered, being the substitution unity), and these 12 substitutions form a group. I omit the details of the proof; it will be sufficient to give the square obtained by means of the several routes, or substitutions, performed upon the primitive arrangement $abcdefghijkl$, and the cyclical expressions of the substitutions themselves: it will be observed that the substitutions are unity, 3 substitutions of the order (or index) 2, and 8 substitutions of the order (or index) 3.

It may be remarked that the group of 12 is really the group of the 12 positive substitutions upon 4 letters $abcd$, viz. these are 1, abc, acb, abd, adb, acd, adc, bcd, bdc, $ab \cdot cd$, $ac \cdot bd$, $ad \cdot bc$.

Cambridge, 16th May, 1878.

No. 3. The Newton-Fourier imaginary problem.

[From the *American Journal of Mathematics*, t. II. (1879), p. 97.]

The Newtonian method as completed by Fourier, or say the Newton-Fourier method, for the solution of a numerical equation by successive approximations, relates to an equation $f(x) = 0$, with real coefficients, and to the determination of a certain real root thereof a by means of an assumed approximate real value ξ satisfying prescribed conditions: we then, from ξ, derive a nearer approximate value ξ_1 by the formula

$$\xi_1 \ = \ \xi - \frac{f(\xi)}{f'(\xi)},$$

and thence, in like manner, $\xi_2, \xi_3, \xi_4, \ldots$ approximating more and more nearly to the required root a.

In connexion herewith, throwing aside the restrictions as to reality, we have what I call the Newton-Fourier Imaginary Problem, as follows.

Take $f(u)$, a given rational and integral function of u, with real or imaginary coefficients; ξ, a given real or imaginary value, and from this derive ξ_1 by the formula

$$\xi_1 \ = \ \xi - \frac{f(\xi)}{f'(\xi)},$$

and thence $\xi_2, \xi_3, \xi_4, \ldots$ each from the preceding one by the like formula.

A given imaginary quantity $x + iy$ may be represented by a point the coordinates of which are (x, y): the roots of the equation are thus represented by given points

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